Lecture 35

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1 Sequences

1.1 Definitions

A sequence (a_n) is a collection $a_1, a_2, a_3, \ldots$ where the order is important. Alternatively, a sequence is a function $f : \mathbb{N} \to \mathbb{R}$ where $f(n) = a_n$.

1.1.1 Recursive Formula

A sequence can be defined recursively. Example: $x_1 = 1$, $x_{n+1} = \frac{x_n+1}{2}$ (reading from context cues, though handwriting is ambiguous).

1.1.2 Monotonicity

A sequence (a_n) is called:

1. **Increasing** if $a_n \leq a_{n+1}$.
2. **Decreasing** if $a_n \geq a_{n+1}$.
3. **Strictly increasing** if $a_n < a_{n+1}$.
4. **Strictly decreasing** if $a_n > a_{n+1}$.
5. **Monotone** if it is either increasing or decreasing.

1.1.3 Boundedness

We always consider real sequences ($(a_n) \subset \mathbb{R}$).

1. **Bounded above**: $\exists M$ such that $a_n \leq M$.
2. **Bounded below**: $\exists m$ such that $a_n \geq m$.
3. **Bounded**: $\exists M$ such that $|a_n| \leq M$.

1.2 Convergence

1.2.1 Definition

A sequence (a_n) **converges** to $a \in \mathbb{R}$ if:

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall n \geq N, |a_n - a| < \epsilon$$

1.2.2 Neighborhoods

Let $\epsilon > 0$. An ϵ-neighborhood of a is the set:

$$U_\epsilon(a) = \{x \in \mathbb{R} : |x - a| < \epsilon\}$$

Geometric interpretation: The interval $(a - \epsilon, a + \epsilon)$.

Alternative Definition: A sequence (a_n) converges to a if every ϵ-neighborhood of a contains all but a finite number of elements of (a_n).

1.2.3 Notation

- Limit: $\lim_{n\to\infty} a_n = a$
- Arrow: $a_n \to a$
- If (a_n) does not converge to any $a \in \mathbb{R}$, it is **divergent**.

1.2.4 The "Game" of Limits

To prove $a_n \to a$:

1. **Challenger**: Gives an arbitrary $\epsilon > 0$ (error tolerance).
2. **Defender**: Must find an index N (threshold).
3. **Check**: For all indices $n \geq N$, the value a_n must lie inside $(a-\epsilon, a+\epsilon)$.

2 Examples

2.1 Example 1: Basic Null Sequence

$$\lim_{n\to\infty} \frac{1}{n} = 0$$

Proof: We must show: $\forall \epsilon > 0, \exists N, \forall n \geq N, |\frac{1}{n} - 0| < \epsilon$. Analysis:

$$\frac{1}{n} < \epsilon \iff n > \frac{1}{\epsilon}$$

Let $\epsilon > 0$. Choose $N > \frac{1}{\epsilon}$ (e.g., if $\epsilon = 10^{-6}$, take $N = 10^6 + 1$). Then for all $n \geq N$, $|\frac{1}{n}| < \epsilon$.

2.2 Example 2: Rational Sequence

$$\lim_{n\to\infty} \frac{n}{n+2} = 1$$

Proof: We must ensure $|\frac{n}{n+2} - 1| < \epsilon$. Simplify the expression:

$$\left|\frac{n}{n+2} - 1\right| = \left|\frac{n-(n+2)}{n+2}\right| = \left|\frac{-2}{n+2}\right| = \frac{2}{n+2}$$

Inequality check:

$$\frac{2}{n+2} < \epsilon \iff n+2 > \frac{2}{\epsilon} \iff n > \frac{2}{\epsilon} - 2$$

Let $\epsilon > 0$. Take $N > \frac{2}{\epsilon} - 2$. Then for all $n \geq N$, $|\frac{n}{n+2} - 1| < \epsilon$.

2.3 Example 3: Quadratic Rational Sequence

$$\lim_{n\to\infty} \frac{2n^2+3}{n^2+n+1} = 2$$

Proof: We examine the difference:

$$\left|\frac{2n^2+3}{n^2+n+1} - 2\right| = \left|\frac{2n^2+3-2(n^2+n+1)}{n^2+n+1}\right|$$

$$= \left|\frac{2n^2+3-2n^2-2n-2}{n^2+n+1}\right| = \left|\frac{1-2n}{n^2+n+1}\right| = \frac{2n-1}{n^2+n+1}$$

We estimate with a simpler expression (finding an upper bound):

$$\frac{2n-1}{n^2+n+1} < \frac{2n}{n^2} = \frac{2}{n}$$

Goal: $\frac{2}{n} < \epsilon$. Let $\epsilon > 0$. Take $N > \frac{2}{\epsilon}$. Then for $n \geq N$:

$$\left|\frac{2n^2+3}{n^2+n+1} - 2\right| < \frac{2}{n} \leq \frac{2}{N} < \epsilon$$

2.4 Example 4: Divergence

Show that $(-1)^n$ is divergent. **Proof (by contradiction):** Suppose $(-1)^n \to a$. Then $\forall \epsilon > 0, \exists N$ such that $\forall n \geq N, |(-1)^n - a| < \epsilon$.

- If n is odd, $(-1)^n = -1$, so $|-1-a| < \epsilon$.
- If n is even, $(-1)^n = 1$, so $|1-a| < \epsilon$.

Using the Triangle Inequality:

$$2 = |1-(-1)| = |1-a+a-(-1)| \leq |1-a| + |a-(-1)| < \epsilon + \epsilon = 2\epsilon$$

If we choose $\epsilon = \frac{1}{2}$ (or any $\epsilon \leq 1$), we get $2 < 1$ (or $2 < 2\epsilon$), which is a contradiction.

3 Theorems

3.1 Uniqueness of Limit

Theorem: If a sequence (a_n) converges, then its limit is unique. **Proof:** Suppose $a_n \to a$ and $a_n \to b$ with $a \neq b$. For any $\epsilon > 0$, there exist N_1, N_2 such that:

- $\forall n \geq N_1, |a_n - a| < \epsilon$
- $\forall n \geq N_2, |a_n - b| < \epsilon$

Let $N = \max\{N_1, N_2\}$. For $n \geq N$:

$$|a - b| = |a - a_n + a_n - b| \leq |a - a_n| + |a_n - b| < \epsilon + \epsilon = 2\epsilon$$

Since this must hold for any ϵ, choose $\epsilon = \frac{|a-b|}{2}$ (or smaller). Then $|a - b| < |a - b|$, a contradiction.

3.2 Boundedness of Convergent Sequences

Theorem: Every convergent sequence is bounded. **Proof:** Let $a_n \to a$. Take a specific ϵ, say $\epsilon = 1$. There exists N such that for all $n \geq N$:

$$|a_n - a| < 1 \implies a - 1 < a_n < a + 1$$

This bounds the "tail" of the sequence. For the initial finite terms $a_1, a_2, \ldots, a_{N-1}$, let:

$$M = \max\{|a_1|, |a_2|, \ldots, |a_{N-1}|, |a| + 1\}$$

Then $|a_n| \leq M$ for all n.